

CUSTOMER SUPPORT TICKET ANALYTICS

Business Insight & Recommendation Report

1. Executive Summary

This report answers the nine business questions in BRD Section 8.

- **Technical shows the most concentrated risk indicators:** Technical shows the most concentrated risk indicators: it accounts for 32% of total volume, represents 46.2% of the open backlog, has the second-slowest resolution time (4,982 min), and the lowest CSAT (3.85).
- **Resolution time differs far more by priority than first-response time does:** Low-priority tickets take approximately 12× longer to resolve than Urgent tickets (6,544 vs. 541 min). A similarly large gap is also observed in first-response time (303 vs. 25 min), so both response and resolution times differ substantially by priority.
- **CSAT does not vary meaningfully with response speed:** average CSAT is flat (3.90–4.03) regardless of how fast the first reply came, in the tickets where CSAT was captured.
- **Technical had the highest number of reopened tickets (84):** It suggesting a potential first-contact-resolution issue that should be investigated. However, the dataset does not record contact reasons or the number of customer touches, so this cannot be confirmed.
- **CSAT coverage is only 30.65%:** (1,226/4,000 tickets), so all CSAT-related findings should be read as directional, not conclusive.

2. Findings by Business Question

BQ / FR	Question	Key data (observed)	Interpretation
BQ-01 FR-03	Top issues by volume?	Top subject: API rate limit question, 60 tickets. Top-10 counts (56–60 each) sum to 577 of 4,000 total tickets.	Top-10 subjects account for ~14.4% of volume. The remaining ~85.6% is spread across other subjects.
BQ-02 FR-02	Highest-work load queues?	Technical 1,282 (32%); Billing 960 (24%); Account 647 (16%); Onboarding 586 (15%); Integrations 525 (13%).	Technical alone accounts for roughly a third of all demand, the largest single concentration of volume.
BQ-03 FR-02	Highest-volume channels?	Email 1,368 (34%); Chat 1,191 (30%); In-app 952 (24%); Phone 489 (12%).	Email + Chat together make up 64% of volume.
BQ-04 FR-04	Slowest first response?	By queue: Integrations 171 min (slowest) vs. Onboarding 130 min (fastest). By priority: Low 303 min vs. Urgent 25 min.	Higher-priority tickets receive substantially faster first responses in the observed data. Integrations has the lowest ticket volume

			but the slowest first-response and resolution times, suggesting its workload allocation or process should be investigated further.
BQ-05 FR-05	Longest resolution time?	By priority: Low 6,544 min (~109h) vs. Urgent 541 min (~9h), which is a 12-times gap. By queue: Integrations 5,732 min, Technical 4,982 min slow.	The resolution-time gap by priority is much wider than the response-time gap, a pattern worth further investigation. Integrations and Technical are slow on both response and resolution.
BQ-06 FR-09, FR-08	FRT vs. CSAT relationship?	CSAT flat at 3.90–4.03 across all response-time bands. CSAT coverage 30.65% which 1,226/4,000. By queue: Technical lowest (3.85), Integrations highest (4.24).	In the tickets with CSAT recorded, response speed does not appear to be a strong predictor of satisfaction; the actual drivers of CSAT are not measurable from this dataset and would need further analysis.
BQ-07 FR-07	Reopened tickets?	Overall 285/4,000 (7.13%). Technical is highest at 84 tickets. Integrations is lowest at 36 tickets.	Technical's high reopen rate suggests a potential first-contact-resolution issue that should be investigated. The dataset has no FCR field, contact-reason, or touch-count to confirm this.
BQ-08 FR-06	Open/unresolved tickets?	225 open (5.63%). Technical+Normal = 51 tickets (22.7% of backlog). Top 4 backlog rows are all Technical or Integrations.	The open-ticket backlog is concentrated in the same two queues that are also slowest to respond and resolve.
BQ-09 FR-01	Monthly volume trend?	Jan–Jul 2023: 282–320 tickets/month, no sharp spikes (full 14-month series available in the dashboard).	Volume in the observed months is relatively stable. No seasonal pattern is evident from this partial window.

3. Cross-Cutting Themes

- **Technical underperforms on every metric measured:** it tops volume, ranks 2nd-slowest to resolve, has the lowest CSAT, and holds the largest backlog share. A queue worth prioritizing for closer review.
- **Slow queues appear to be slow for different reasons:** Integrations combines the lowest volume with the slowest response and resolution times; Technical combines the highest volume with among the slowest resolution times. These are different patterns that likely need different investigation, since the dataset cannot confirm whether staffing, process, or ticket complexity is the cause.
- **Priority has a substantial impact on both first-response and resolution times:** Low-priority tickets take approximately 12× longer than Urgent tickets for both first response (303 vs. 25 minutes) and resolution (6,544 vs. 541 minutes). This pattern is consistent with priority-based handling, although the operational rules behind the differences are not available in the dataset.
- **Response speed alone does not appear to explain satisfaction:** CSAT is flat (3.90–4.03) across all response-time bands in the tickets where it was captured. Other factors not present in this dataset would need to be analyzed to understand CSAT drivers.

4. Recommendations

1. **Investigate workload allocation for Technical and Integrations**
Technical drives the most volume and backlog. Integrations has the slowest first response despite the lowest volume. Consider reviewing staffing levels and process for both queues, since the underlying cause cannot be determined from this dataset alone (BQ-02, BQ-04, BQ-08).
2. **Consider introducing a working SLA of approximately 5 business days for Low-priority tickets**
The current average resolution time (~109 hours) is well beyond typical SLA norms. The exact threshold has not been derived from the data and should be validated against business capacity (BQ-05).
3. **Investigate Technical's high reopen quantity (84, the highest among all queues)**
The high number of reopened Technical tickets may warrant a review of common Technical subjects and handling processes as a potential first-contact-resolution issue. However, the dataset does not include contact reasons or number of customer touches, so a specific cause cannot be confirmed (BQ-07).
4. **Increase CSAT survey completion to improve coverage beyond the current 30.65% baseline**
For example, the survey could be triggered immediately after resolution and shortened to encourage completion, strengthening the coverage of future satisfaction analysis (FR-08).
5. **Conduct a follow-on analysis to test whether reopened or multi-touch tickets are associated with lower CSAT**
CSAT remains within a relatively narrow range across the first-response-time bands in the current dataset. Further analysis of reopened and multi-touch tickets could help identify whether these factors are associated with customer satisfaction (BQ-06).

5. Data Quality Notes & Conclusion

- **CSAT coverage is 30.65%:** the excluded 69.35% are omitted (not imputed) from averages, per the BRD's "missing at random" assumption. Therefore, all CSAT-related findings should be read as directional, not conclusive.
- **Agent-level and staffing data are out of scope:** no agent/assignee, shift, or capacity field exists in the source data (BRD Section 5.2), so capacity, coverage, or workload claims cannot be directly verified. Recommendations therefore target queues and processes for further review, not confirmed fixes.
- **No root-cause or contact-reason fields exist:** reopened status is recorded, but the reason for reopening and number of customer contacts are not available. Therefore, first-contact-resolution conclusions remain an interpretation to be tested, not a proven finding.
- **This is a static, historical extract:** findings describe the period from January 2023 to February 2024 and are not a live view of current performance.

Overall, the open backlog represents 5.63% of total ticket volume, with 225 open tickets out of 4,000. The data points to three areas warranting further attention: the Technical queue's high volume, largest backlog share, and lowest CSAT. The substantially longer resolution time for Low-priority tickets; and a potential first-contact-resolution issue indicated by the high number of reopened Technical tickets. Each of the five recommendations above is framed as an action to investigate, test, or consider, consistent with the dataset's scope, rather than as a proven fix. Pursuing these actions would help build the consolidated, evidence-based view of support operations that the BRD identifies as currently missing.